

Multi-UUV Maneuvering Counter-Game for Dynamic Target Scenario Based on Fractional-Order Recurrent Neural Network

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Abstract—In this article, a multi-underwater unmanned vehicle (UUV) maneuvering decision-making algorithm is proposed for a counter-game with a dynamic target scenario. The game is modeled with interval-valued intuitionistic fuzzy rules, and an optimal maneuvering strategy is realized using a fractional-order recurrent neural network (RNN). First, underwater environments with weak connectivity, underwater noise, and dynamic uncertainties are analyzed and incorporated into the interval-valued intuitionistic fuzzy set. Then, the maneuvering decision-making model and the expected return of the multi-UUV countermeasure are designed based on the interval-valued intuitionistic fuzzy rules. Subsequently, to optimize the counter-game maneuvering strategy, a fractional-order RNN is formulated based on the Karush–Kuhn–Tucker optimality conditions. In addition, the existence and uniqueness of the optimal maneuvering solutions as well as the stability of the equilibrium point are discussed. Finally, simulation and experimental results are compared to determine the effectiveness of the proposed algorithm. The influence of the fractional order on the convergence rate and optimization error of the proposed algorithm is also minutely examined.

Index Terms—Decision making, fractional calculus, interval fuzzy game theory, multi-underwater unmanned vehicle (UUV), recurrent neural network (RNN).

I. INTRODUCTION

WITH rapid advancements in underwater unmanned vehicle (UUV) technology, the difficulty and complexity of its applications have advanced significantly. Owing to

limitations, such as limited loading, low hit probability, and a small detection range, a single UUV barely fulfills the requirements of current tasks. Multi-UUVs can cooperate with each other according to the actual underwater situation, investigate, and share resources, and effectively increase the success rate of countermeasures along with the survival rate of single UUVs. Currently, most multi-UUV control research focus on their own strategy optimization, such as formation control, collaborative target tracking, and navigation and positioning, which has advanced significantly [1], [2]. A multi-UUV counter-game is a conflict decision-making process for both sides in the game, that is, a multi-UUV system must consider the strategy of the enemy along with its own strategy [3]. Moreover, the complex underwater environment and multiple uncertainties render the maneuvering decision-making process of the multi-UUV counter-game quite challenging.

Recently, studies on decision-making algorithms for unmanned systems have attracted considerable research attention [4], [5], [6], [7], [8], [9], [10], [11]. Ma et al. [4] proposed a cooperative occupancy decision-making algorithm for multi-unmanned aerial vehicles (UAVs) in beyond-visual-range air combat. The algorithm was presented in a relatively ideal environment where the uncertainties in UAV performance and external environment were not considered. A self-organized offense–defense confrontation decision-making algorithm for a dynamic swarm versus swarm UAVs combat problem has been studied by Xing et al. [5]. However, some prior information was introduced in the corresponding decision-making process, thereby reducing the application scope. Another intelligent self-organized decision-making problem was presented by Zhen et al. for a cooperative search–attack mission of multi-UAV [6]. The employed algorithm divided the global optimization problem into several local ones such that the optimal decision was made based on information exchange among UAVs. Duan et al. [7] analyzed the uncertainty of the target value and damage probability of a multi-UAV system and proposed a multi-UAV countermeasure model in an uncertain attack and defense game environment. Radmanesh et al. [9] studied the dynamic game model of multi-UAV and incorporated uncertain random factors into the model to obtain the Nash equilibrium for uncertain information, which reflected the real confrontation situation. Chai et al. [8] presented a decision-making algorithm based on deep neural networks (DNNs) for the optimal trajectory planning problem of six-degree-of-freedom spacecraft. Moreover,

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a deep-reinforcement-learning-based multi-autonomous underwater vehicles (AUVs) cooperative decision making was proposed by Xu et al. for the offense–defense confrontation problem [10]. The above studies demont the superior decision-making performance achieved using intelligent learning methods, such as DNNs and deep reinforcement learning. The related intelligent learning methods may show extra potential in cooperative decision-making problems for swarm underwater vehicles. However, underwater communication and environmental uncertainty, which are critical constraints in such problems, must be well-presented first.

Game theory is suitable for highlighting the bilateral conflicting nature of the maneuvering decision-making problem of cooperative countermeasures. In the literature, the application of game theory to model the mechanism of a multi-unmanned system decision-making process has already been explored [12]. However, studies on underwater counter-game models are lacking. Owing to the unique marine communication and environmental conditions, the multi-UUV maneuvering decision-making process is more complex than other unmanned vehicles [13]. The uncertainty in sonar and other factors may result in low efficiency, severe delays, and other connectivity constraints linked to weak underwater communication, which cause the information of underwater countermeasures to exhibit fuzzy characteristics [14]. These communication obstacles result in severe constraints for optimizing the multi-UUV countermeasures. Therefore, establishing a counter-game model that includes underwater features and generates game benefits in underwater conditions remains to be achieved. In this study, an interval-valued intuitionistic fuzzy game theory is used to model the maneuvering decision-making process of a multi-UUV to deploy underwater countermeasures for a dynamic target scenario. Classical game theory only considers clear and specific payments [15]. However, in a marine environment, owing to poor communication conditions and environmental uncertainties, the information-interaction and perception abilities are weak [14]. The situational information obtained by the multi-UUV system is mostly fuzzy information and always uncertain. The interval-valued intuitionistic fuzzy set can better describe the uncertainties of the game participant attributes compared to other methods. Therefore, it is challenging and suitable to incorporate the characteristics of underwater conditions in the counter-game model for the maneuvering decision-making process [16].

Post modeling, the next challenge for multi-UUV maneuvering in a counter-game for a dynamic target scenario is the real-time optimization efficiency. Recently, some related remarkable results have been published in the field of strategy optimization. For example, multiobjective trajectory optimization methods of unmanned systems are provided in [17], [18], and [19], based on adaptive differential evolution (DE) and modified game theory. Owing to the calculation complexity of a dynamic decision-making process of multi-UUV, the optimization efficiency may severely affect the selection of an optimal strategy for multi-UUV countermeasures. Therefore, we propose a fractional-order recurrent neural network (RNN) model to improve the optimization efficiency.

Developing an RNN model to solve a constrained optimization problem was previously proposed by Hopfield and Tank [20]. In [21], an RNN model was applied to develop new strategies to solve fuzzy matrix games. Owing to its long-term memory and hereditary property [22], fractional-order neural networks have been deemed a major improvement over integer-order neural networks [23]. The properties of fractional-order neural networks, such as stability, synchronization, and bifurcation, have been widely investigated [24], [25], [26]. In particular, some Mittag-Leffler stability criteria of fractional-order neural networks are markedly helpful in the dynamic analysis and application in optimization problems [27], [28], [29]; In addition, they provide a theoretical basis for convergence in the present study. Fractional-order RNNs may exhibit outstanding performance in optimizing multidimensional parameters compared to traditional optimization algorithms such as particle swarm optimization (PSO) and the genetic algorithm. Optimization parameters with more dimensions always generate inefficient iterations in PSO or the genetic algorithm. However, the convergence rate of a fractional-order RNN is mainly dependent on its dynamics; therefore, it has great potential in multidimensional optimization and decision-making problems. Moreover, the order of a fractional-order RNN can effectively tune and balance the convergence rate and optimization error. Thus, the real-time multi-UUV maneuvering decision-making problem can be solved more effectively by constructing a fractional-order RNN of an appropriate order.

Overall, the main contributions of the multi-UUV maneuvering counter-game for the dynamic target scenario proposed herein can be stated as follows.

- 1) *Accurate and Detailed Modeling*: Complex underwater communication conditions as well as unpredictable environmental and equipment uncertainties were represented in the counter-game modeling process using interval-valued intuitionistic fuzzy rules.
- 2) *Reasonable Strategy Optimization*: A fractional-order RNN, which exhibits a convergence rate that mainly depends on its dynamics, was proposed for the multidimensional strategy optimization in the game process. By integrating the interval-valued intuitionistic fuzzy game and fractional-order RNN, we were able to obtain a good adversarial strategy of the multi-UUV countermeasure.
- 3) *High Efficiency of Real-Time Underwater Decision Making*: The order of a fractional-order RNN can affect the convergence rate and optimization error. The order can be tuned to satisfy the requirements of real-time strategy optimization, thereby increasing the efficiency of real-time multi-UUV maneuvering decision making.

The remainder of this article is organized as follows. In Section II, several preliminaries of interval-valued intuitionistic fuzzy game theory are provided. A detailed evaluation of the maneuvering attributes of the multi-UUV system is presented in Section III. Section IV presents the maneuver decision-making model that we prepared using interval-valued intuitionistic fuzzy game theory. The strategy optimization method based on the fractional RNN model is presented in

Section V. Simulation and experimental results obtained to verify the proposed multi-UUV maneuver decision-making algorithm for the dynamic target scenario are presented in Sections V and VII, respectively. Finally, conclusion are provided in Section VIII.

II. PRELIMINARIES

A. Interval Intuitionistic Fuzzy Game Theory

Both fuzzy sets and interval fuzzy sets use the membership degree or membership degree interval of a single scale to express the two opposite states of fuzziness, fuzzy phenomenon, or fuzzy concept. However, in the marine environment, the membership degree of elements belonging to a certain set is often determined without absolute assurance. That is, the membership degree contains a certain amount of hesitation or uncertainty. It is possible that elements will exhibit both subordination and insubordination to a fuzzy set, and they may simultaneously show a certain degree of hesitation. To describe such complexity and uncertainty, Atanassov and Garov [16] proposed the concept of an interval intuitionistic fuzzy set based on an intuitionistic fuzzy set. The membership and nonmembership of an intuitionistic fuzzy set are expressed in the form of an interval number. We use the interval intuitionistic fuzzy game theory to describe the decision-making process of multi-UUV countermeasures.

Definition 1: Let X be a nonempty field. Moreover, $\mu_A : X \rightarrow I_{[0,1]}$ and $\nu_A : X \rightarrow I_{[0,1]}$ are two interval value mappings on X , and $I_{[0,1]}$ represents the unit interval $[0, 1]$. If the following is satisfied: $0 \leq \sup\{\mu_A(x)\} + \sup\{\nu_A(x)\} \leq 1$ then

$$A = \{ \langle x, [\mu_A^L(x), \mu_A^U(x)], [\nu_A^L(x), \nu_A^U(x)] \rangle | x \in X \} \quad (1)$$

is defined as an interval intuitionistic fuzzy set, where $[\mu_A^L(x), \mu_A^U(x)]$ and $[\nu_A^L(x), \nu_A^U(x)]$ denote the membership and nonmembership intervals, respectively, of A , where $\mu_A^L(x) \in [0, 1]$, $\mu_A^U(x) \in [0, 1]$, $\nu_A^L(x) \in [0, 1]$, $\nu_A^U(x) \in [0, 1]$, $\mu_A^U(x) + \nu_A^U(x) \leq 1$. For simplification, we refer to

$$\alpha(x) = ([\mu^L(x), \mu^U(x)], [\nu^L(x), \nu^U(x)]) \quad (2)$$

as an interval intuitionistic fuzzy number.

Definition 2: For the interval intuitionistic fuzzy set A on field X , $\pi_A = [1 - \mu_A^U(x) - \nu_A^U(x), 1 - \mu_A^L(x) - \nu_A^L(x)]$ is defined as the interval hesitancy degree of element x on field X subordinated to A .

Definition 3: Let $\alpha_1 \sim$ and $\alpha_2 \sim$ be two interval intuitionistic fuzzy numbers, $\alpha_1 \sim = ([a_1, b_1], [c_1, d_1])$ and $\alpha_2 \sim = ([a_2, b_2], [c_2, d_2])$. The basic operation rules are then defined as follows:

$$\begin{aligned} \tilde{\alpha}_1 \oplus \tilde{\alpha}_2 &= ([a_1 + a_2 - a_1 a_2, b_1 + b_2 - b_1 b_2], [c_1 c_2, d_1 d_2]) \\ \lambda \tilde{\alpha} &= ([1 - (1 - a)^\lambda, 1 - (1 - b)^\lambda], [c^\lambda, d^\lambda]), \lambda > 0. \end{aligned} \quad (3)$$

B. Fractional-Order RNN

RNN is a common deep-learning algorithm widely used in applications, such as language processing, time-series

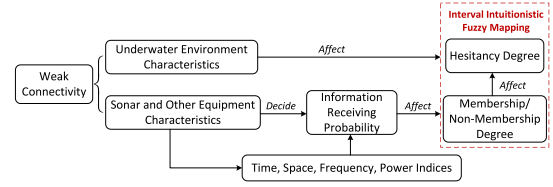


Fig. 1. Fuzzy mapping of the weak connectivity constraint.

prediction, and optimization. Recently, fractional calculus is always used to describe the characteristics of the long-term memory in an RNN model. Herein, a fractional-order RNN with the Caputo differential [25] is introduced as

$${}^C D_t^\gamma x_i(t) = -c_i x_i(t) + \sum_{j=1}^n b_{ij} f_j(x_j(t)) + I_i, \quad i = 1, 2, \dots, n$$

where $0 < \gamma < 2$ denotes the fractional order of the Caputo differential, n denotes the number of units in the neural network, $x_i(t)$ denotes the state of the i th unit, $f_j(x_j(t))$ denotes the j th neuronal activation function, c_i denotes the self-regulating coefficient of the i th neuron, b_{ij} denotes the weight coefficients between the i th and j th neurons, and I_i denotes the i th external input.

To simplify the above fractional-order RNN model, the equivalent equation is given by

$${}^C D_t^\gamma x(t) = F(x(t)) \quad (4)$$

where $x(t) = (x_1(t), x_2(t), \dots, x_n(t))^T$ and $F(x(t)) = (F_1(x(t)), F_2(x(t)), \dots, F_n(x(t)))^T$ when satisfying

$$F_i(x(t)) = -c_i x_i(t) + \sum_{j=1}^n b_{ij} f_j(x_j(t)) + I_i.$$

In Section V, a fractional-order RNN model is developed to solve the strategy optimization in the game process. The corresponding principles and steps are also presented.

III. MANEUVER ATTRIBUTE EVALUATION OF MULTI-UUV SYSTEM

A. Interval Intuitionistic Fuzzy Mapping of Weak Connectivity Constraints

Weak connectivity constraints mainly stem from underwater environments, such as the characteristics of sonar and other equipment, ocean reverberation, and environmental noise [30]. In the interval intuitionistic fuzzy game-based decision-making process presented in this work, the receiving probability of underwater information is mainly determined by the characteristics of sonar and other equipment; hence, it will affect the membership and nonmembership degrees of the countermeasure maneuvering attributes. The corresponding hesitance degree is affected by the underwater environment characteristics; that is, an increase in environmental uncertainty will lead to an increase in the hesitancy degree of the attributes. A sketch of the fuzzy mapping is depicted in Fig. 1.

The characteristics of sonar and other equipment are mainly reflected in the time, space, frequency, and power indices, which influence the information transmission as follows.

- 1) *Time Index*: This indicates that the multi-UUV countermeasures should be implemented within the sonar action time, which can be measured by the countermeasure opportunity factor f_t

$$f_t = \begin{cases} \frac{t_{s_2}-t_D}{t_{s_2}-t_{s_1}}, & t_{s_1} \leq t_D \leq t_{s_2} \\ 0, & t_D > t_{s_2} \text{ or } t_D < t_{s_1} \end{cases} \quad (5)$$

where t_{s_1} and t_{s_2} represent the start and end times of the sonar operation, respectively; t_D denotes the start time of the underwater countermeasure; and f_t indicates the extent to which the receiving signal is affected by the sonar characteristics in the time domain. A larger f_t value results in better information transmission and reception.

- 2) *Space Index*: This means that the power of our countermeasure equipment should cover the enemy multi-UUV system location, which can be measured by the space coverage factor f_w

$$f_w = \begin{cases} 1, & w_s \in P(w_D, r_1) \\ \frac{r_2-r}{r_2-r_1}, & w_s \in P(w_D, r_2), w_s \notin P(w_D, r_1) \\ 0, & w_s \notin P(w_D, r_2) \end{cases} \quad (6)$$

where w_D represents the location of our multi-UUV system, w_s is the location of the enemy multi-UUV system, and $P(w_D, r)$ denotes the countermeasure space formed with r as the action radius, such that r is the linear distance between w_s and w_D . $P(w_D, r_1)$ indicates that a strong attack can be performed within the scope of r_1 ; $P(w_D, r_2)$ indicates that the attack effect can be ignored outside the scope of r_2 ; and r_1 and r_2 are determined by the characteristics of the sonar equipment and should satisfy $P(w_D, r_1) \in P(w_D, r_2)$.

- 3) *Frequency Index*: This indicates that the signal frequency of the multi-UUV countermeasures must be within the detection range of the sonar equipment, which can be measured by the countermeasure frequency factor f_ω

$$f_\omega = \begin{cases} 1 - \frac{\min(\omega_{D_2}, \omega_{S_2}) - \max(\omega_{D_1}, \omega_{S_1})}{\omega_{S_2} - \omega_{S_1}}, & \omega_{S_1} \leq \omega_{D_2} \\ & \omega_{S_2} \geq \omega_{D_1} \\ 1, & \omega_{S_1} > \omega_{D_2} \cup \omega_{S_2} < \omega_{D_1} \end{cases} \quad (7)$$

where $(\omega_{S_1}, \omega_{S_2})$ denotes the operation frequency range of our multi-UUV sonar equipment and $(\omega_{D_1}, \omega_{D_2})$ denotes the operation frequency range of that of the enemy. A larger f_ω value results in a stronger attack that the enemy is likely to receive.

- 4) *Power Index*: This represents the power ratio of the countermeasure signal to the detected target signal; it can be measured by comparing the power between our counter signal and the target signal

$$P_{DS} = E_D - E_s. \quad (8)$$

Here, E_D denotes the transmitted power intensity of the sonar equipment and E_s denotes the power intensity loss for detecting reflected or radiated signals. Because the sonar equations used by different countermeasures and those of active and passive detections

differ, using the correct sonar equations and the corresponding E_D , E_s according to the actual countermeasure methods is necessary.

The time, space, frequency, and power indices reflect the underwater communication situations during the countermeasure process. The space and frequency indices represent the detection ability of our communication signal, similar to the power index. Therefore, they can be represented by the change in the signal-to-noise ratio (SNR) at the output

$$f_\theta = f_w f_\omega P_{DS}. \quad (9)$$

Subsequently, the probability coefficient of the countermeasure information power transmission can be expressed as follows:

$$f_P = P_d(P_f, 10^{0.1(-10 \log(f_\theta/K_D) + T_{0.1})}) \quad (10)$$

where K_D represents the sonar suppression coefficient; $P_d(P_f, T_{PDS})$ denotes the detection probability function of single pulse linear detection; $T_{0.1}$ denotes the signal detection factor when the false alarm probability P_f gives $P_d = 0.1$; f_P denotes the probability that our counter information power can be received, in which $P_d(P_f, T_{PDS}) = e^{-T_{PDS}} \int_{\sqrt{-2 \ln(P_f)}}^{+\infty} x e^{-(x^2/2)} I_0(x \sqrt{2 T_{PDS}}) dx$ denotes the SNR of sonar detection, and $I_0(x)$ denotes a zero-order modified Bessel function of x , $I_0(x) = \sum_{n=0}^{\infty} ([x^{2n}]/[2^{2n} \cdot n! \cdot n!])$.

Therefore, the total sonar and equipment information receiving probability P can be expressed as follows:

$$P = f_P f_\theta, \quad P \in [0, 1]. \quad (11)$$

For the interval intuitionistic fuzzy set $A = \langle x, [\mu_A^L(x), \mu_A^U(x)], [v_A^L(x), v_A^U(x)] \rangle | x \in X$ introduced in the previous section, the interval intuitionistic scalar $\pi_A = [1 - \mu_A^U(x) - v_A^U(x), 1 - \mu_A^L(x) - v_A^L(x)]$ represents the impact of the underwater environment uncertainties on the hesitancy degree of the countermeasure maneuvering attributes. Further, considering the influence of the sonar and equipment characteristics on the membership and non-membership of the countermeasure maneuver attributes, we introduce the probability function P of the sonar and equipment information reception, resulting in the following intuitionistic fuzzy sets with an underwater weak connectivity constraint:

$$A = \{ \langle x, P[\mu_A^L(x), \mu_A^U(x)], P[v_A^L(x), v_A^U(x)] \rangle | x \in X \} \quad (12)$$

where the interval intuitionistic scalar is updated as $\pi_A = [1 - P\mu_A^U(x) - Pv_A^U(x), 1 - P\mu_A^L(x) - Pv_A^L(x)]$. The updated intuitionistic fuzzy mapping not only includes the influence of the underwater environment uncertainties on the hesitancy degree of the countermeasure maneuvering attributes but also represents the influence of the information receiving probability of the sonar and equipment on the membership, nonmembership, and hesitancy degrees.

B. Interval Intuitionistic Fuzzy Game of Multi-UUV for Dynamic Target Scenario

The dynamic target scenario's multi-UUV countermeasure game includes five main elements.

- 1) *Participants in the Game*: The participants in this multi-UUV counter-game are our multi-UUV system and the enemy multi-UUV system.
- 2) *Maneuvering Strategy Profile*: In every dynamic counter-game stage, each participant chooses a maneuvering strategy profile to ensure that its side achieves the maximum benefit or pays the minimum cost.
- 3) *Pure Strategy and Mixed Strategy*: A pure strategy is unique and definite. However, in a mixed strategy, the participants select a strategy profile with a certain probability distribution. When the selected probability of a strategy is 1, it is a pure strategy.
- 4) *Payment or Payoff Value*: This refers to the income value obtained by both sides of the game according to their own selected strategies. The payment function or payoff matrix of the game is obtained from evaluating the countermeasure situation.
- 5) *Equilibrium of Game*: This refers to the optimal mixed strategy profile of all participants in a zero-sum game. Based on the classical theory of Nash equilibrium, there exists a mixed strategy pair (x^*, y^*) in an ideal zero-sum game $(W \times E)$, such that

$$E_W(x, y^*) \leq E_W(x^*, y^*) \leq E_W(x^*, y) \quad \forall x \in S_W \quad \forall y \in S_E$$

where S_W and S_E denote the strategy sets of W and E , respectively; E_W and E_E are the profits of W and E , respectively, that is, $E_W = -E_E$. The above (x^*, y^*) is called the equilibrium of game. If a participant W abandons this equilibrium state x^* and the other participant E maintains the equilibrium strategy y^* , the profit of W may decrease and the profit of E may increase.

In the multi-UUV countermeasures process, the multi-UUVs from both sides are regarded as two participants. The finite process for a dynamic target scenario game can be divided into k stages, so that the dynamic expansion model of the game can be described as $G = \{N, S, U\}$, where $N = 1, 2$ represents our multi-UUV and the enemy multi-UUV in the game; $S = \{s_{1i}^k, s_{2j}^k\}$ are the maneuvering strategy spaces of the participants; s_{1i}^k refers to our multi-UUV selecting the i th maneuvering strategy at the k th stage of the game; s_{2j}^k means that the enemy multi-UUV selects the j th maneuvering strategy at the k th stage of the game; and $U = \{u_1(s^{k,j}), u_2(s^{k,j})\}$ is the payment of each participant corresponding to each maneuvering strategy that the participant may select.

IV. MANEUVER DECISION-MAKING MODEL USING INTERVAL INTUITIONISTIC FUZZY GAME THEORY

A. Interval Intuitionistic Fuzzy Evaluation of Multi-UUV Maneuvering Strategies

Maneuvering decision making is a critical problem in a multi-UUV noncooperative counter-game, which is the basic action of each game step. To establish the payment matrices of both multi-UUV systems, maneuvering strategy evaluation should first proceed according to the actual situation information. The countermeasure trajectory of the multi-UUVs can be regarded as a combination of each maneuvering

TABLE I
RELATIONSHIP BETWEEN THE FUZZY LANGUAGE AND FUZZY SETS

Fuzzy language	Intuitionistic fuzzy sets	Threat level
Extremely Strong	(1, 0)	1
Strong	(0.85, 0.05)	2
Moderately Strong	(0.75, 0.1)	3
Medium	(0.6, 0.25)	4
Moderately Weak	(0.5, 0.35)	5
Weak	(0.35, 0.6)	6
Extremely Weak	(0.1, 0.85)	7

TABLE II
IMPORTANCE LEVEL OF DIFFERENT ATTRIBUTES

Relative Degree	Extremely Strong	Moderately Strong	Strong	Medium Strong	Equivalent
Value	2	1.75	1.5	1.25	1.0

action. The maneuvering actions may exhibit certain limitations according to the maneuver features of different UUVs. Here, the two multi-UUV systems in the underwater countermeasures are denoted as “W” and “E” for simplicity. For every i th UUV, the maneuvering strategy sets S_{Wi} and S_{Ei} for these systems can be defined as

$$\begin{cases} S_{Wi} = \{w_{i1}, \dots, w_{i7}\} \\ S_{Ei} = \{e_{i1}, \dots, e_{i7}\} \end{cases} \quad (13)$$

where several maneuver strategies are listed as follows. w_{i1} and e_{i1} denote “maintain the pace,” w_{i2} and e_{i2} denote “speed up,” w_{i3} and e_{i3} denote “speed down,” w_{i4} and e_{i4} denote “left turn,” w_{i5} and e_{i5} denote “right turn,” w_{i6} and e_{i6} denote “pitch up,” and w_{i7} and e_{i7} denote “pitch down.”

Four maneuvering attributes are considered

$$R = \{R_{\text{dis}}, R_{\text{vel}}, R_{\text{ang}}, R_{\text{dng}}\} \quad (14)$$

where R is the maneuver attribute set, R_{dis} represents the distance factor, R_{vel} represents the velocity factor, R_{ang} represents the deflection angle, and R_{dng} represents the depression angle.

The main difference between multi-UUV countermeasures and other autonomous robot countermeasures is the information transmission mode. The communication restrictions and marine environmental uncertainties result in the fuzzy property of underwater information, rendering the accurate quantification of the threat level of both multi-UUVs in the decision-making process difficult. Therefore, herein, the threat level of each attribute is categorized into seven sub-levels through the interval intuitionistic fuzzy language. This is because the interval intuitionistic fuzzy set can measure the degree of fuzziness of the original information more comprehensively; thus, the fuzzy language is transformed into intuitionistic fuzzy sets. Table I lists the relationship between the fuzzy language and interval intuitionistic fuzzy sets.

The importance ranking of the maneuvering attributes is obtained based on the analytic hierarchy process [31]. The value of the importance level I_{ij} of the i th maneuvering attribute factor R_i relative to the j th one R_j is obtained using the arithmetic mean method, as indicated in Table II, which may change according to practical problems.

Subsequently, the importance level matrix I can be obtained as follows:

$$I = \begin{bmatrix} 1 & I_{12} & I_{13} & I_{14} \\ I_{21} & 1 & I_{23} & I_{24} \\ I_{31} & I_{32} & 1 & I_{34} \\ I_{41} & I_{42} & I_{43} & 1 \end{bmatrix} \quad (15)$$

where I_{ji} denotes the reciprocal of I_{ij} according to the importance level definition.

The threat weight t_k ($k = 1, 2, 3, 4$) of each attribute can be obtained as follows:

$$t_k = \sum_{j=1}^4 I_{kj} / \sum_{k=1}^4 \sum_{j=1}^4 I_{kj}. \quad (16)$$

As noted in the previous section, two types of multi-UUV counter-game models exist: the pure strategy and mixed strategy models. In the countermeasures process, both multi-UUVs should select their maneuvering strategy according to the actual situation information, and then calculate the payoff matrices. The dimensions of the maneuvering strategy sets S_W and S_E are both set as q . The interval intuitionistic fuzzy set $f_{mn} = ([u_{mn}^L, u_{mn}^U], [v_{mn}^L, v_{mn}^U])$ ($m, n = 1, 2, \dots, q$) can be obtained according to Table 1 to quantitatively evaluate the selected strategy, where u_{mn} and v_{mn} denote the membership and nonmembership degrees, respectively [32]. Therefore, the fuzzy evaluation matrix F under the attributes R_{dis} , R_{vel} , R_{ang} , and R_{dng} of the multi-UUV “W” is expressed as follows:

$$F = \begin{matrix} & e_1 & e_2 & \cdots & e_n \\ \begin{matrix} w_1 \\ \vdots \\ w_m \end{matrix} & \begin{bmatrix} f_{11} & f_{12} & \cdots & f_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ f_{m1} & f_{m2} & \cdots & f_{mn} \end{bmatrix} \end{matrix}. \quad (17)$$

Definition 4: For the intuitionistic fuzzy set $f_{mn} = ([u_{mn}^L, u_{mn}^U], [v_{mn}^L, v_{mn}^U])$, ($m, n = 1, 2, \dots, q$), the weighted arithmetic integration factor (IFWA_{*t*}) is defined as

$$\text{IFWA}_t = \sum_{k=1}^4 t_k F_k = \left(1 - \prod_{k=1}^4 (1 - u_k)^{t_k}, \sum_{k=1}^4 v_k^{t_k} \right) \quad (18)$$

where q denotes the dimension of the maneuvering strategy sets S_W and S_E , and t_k denotes the threat weight that satisfies $0 \leq t_k \leq 1$, $\sum_{k=1}^4 t_k = 1$. Therefore, the interval intuitionistic fuzzy payoff matrix $[G^L, G^U]$ can be obtained as follows:

$$G^L = \begin{bmatrix} g_{11}^L & \cdots & g_{1n}^L \\ \vdots & \ddots & \vdots \\ g_{m1}^L & \cdots & g_{mn}^L \end{bmatrix}, G^U = \begin{bmatrix} g_{11}^U & \cdots & g_{1n}^U \\ \vdots & \ddots & \vdots \\ g_{m1}^U & \cdots & g_{mn}^U \end{bmatrix} \quad (19)$$

where $g_{mn}^L = (u_{mn}^L, v_{mn}^L)$, $u_{mn}^L = 1 - \prod_{i=1}^4 (1 - u_{mn}^{(i)L})^{t_i}$, $v_{mn}^L = \prod_{i=1}^4 (v_{mn}^{(i)L})^{t_i}$, $g_{mn}^U = (u_{mn}^U, v_{mn}^U)$, $u_{mn}^U = 1 - \prod_{i=1}^4 (1 - u_{mn}^{(i)U})^{t_i}$, $v_{mn}^U = \prod_{i=1}^4 (v_{mn}^{(i)U})^{t_i}$ ($m, n = 1, 2, \dots, q$).

B. Maneuvering Decision-Making Model of Multi-UUV Counter-Game

In multi-UUV countermeasures, a real-time change in the countermeasure situation information makes it difficult for both sides to select the other's strategy in advance. The

main characteristic of the proposed maneuvering decision-making algorithm is that the strategy of each dynamic stage is determined according to the real-time profits adopted by our multi-UUV and the enemy multi-UUV. Although the underwater situation information is incomplete owing to the underwater communication and marine environmental uncertainties, the optimal strategy can still be found with the maximum probability. The multi-UUV countermeasure game with uncertain underwater information is essentially a two-person zero-sum game. The fuzzy payoff matrix F was obtained in the previous section. In this case, the pure strategies of the participants in the countermeasure game “W” and “E” are $w_m \in S_W$ and $e_n \in S_E$ with probabilities p_m^w and p_n^e , respectively. We denote $w = (w_1, w_2, \dots, w_m)^T$ and $e = (e_1, e_2, \dots, e_n)^T$ as the mixed strategies of “W” and “E,” respectively. Then

$$\begin{aligned} P^W &= \left\{ p^w \mid p^w = (p_1^w, p_2^w, \dots, p_m^w)^T \in R^n, \dots, \right. \\ &\quad \left. \sum_{i=1}^m p_i^w = 1, p_i^w \geq 0, i = 1, 2, \dots, m \right\} \\ P^E &= \left\{ p^e \mid p^e = (p_1^e, p_2^e, \dots, p_n^e)^T \in R^n, \dots, \right. \\ &\quad \left. \sum_{j=1}^n p_j^e = 1, p_j^e \geq 0, j = 1, 2, \dots, n \right\} \end{aligned} \quad (20)$$

are the mixed strategy spaces of “W” and “E.” Therefore, $\Gamma = (W, P^W, E, P^E; G)$ is the interval intuitionistic fuzzy two-person noncooperative zero-sum game.

Definition 5: Under the mixed strategy (P^W, P^E) , the expected return $\bar{E}_W^L, \bar{E}_W^U$ of participant “W” can be expressed as follows:

$$\begin{aligned} \bar{E}_W^L(p^w, p^e) &= p^{wT} G p^e \\ &= \left(1 - \prod_{i=1}^m \prod_{j=1}^n (1 - u_{ij}^L)^{p_i^w p_j^e}, \prod_{i=1}^m \prod_{j=1}^n v_{ij}^{L p_i^w p_j^e} \right) \\ \bar{E}_W^U(p^w, p^e) &= p^{wT} G p^e \\ &= \left(1 - \prod_{i=1}^m \prod_{j=1}^n (1 - u_{ij}^U)^{p_i^w p_j^e}, \prod_{i=1}^m \prod_{j=1}^n v_{ij}^{U p_i^w p_j^e} \right) \end{aligned}$$

according to the algorithms of fuzzy sets [16]. Further, the expected return $[\Delta \bar{E}_W^L, \Delta \bar{E}_W^U]$ of participant “W” is

$$\begin{aligned} \Delta \bar{E}_W^L(p^w, p^e) &= 1 - \prod_{i=1}^m \prod_{j=1}^n (1 - u_{ij}^L)^{p_i^w p_j^e} - \prod_{i=1}^m \prod_{j=1}^n v_{ij}^{L p_i^w p_j^e} \\ \Delta \bar{E}_W^U(p^w, p^e) &= 1 - \prod_{i=1}^m \prod_{j=1}^n (1 - u_{ij}^U)^{p_i^w p_j^e} - \prod_{i=1}^m \prod_{j=1}^n v_{ij}^{U p_i^w p_j^e}. \end{aligned} \quad (21)$$

Similarly, the expected return $[\Delta \bar{E}_E^L, \Delta \bar{E}_E^U]$ of participant “E” can be calculated using Definition 5. Thus, the expected return of the interval intuitionistic fuzzy game between “W” and “E” is

$$[\Delta \bar{E}^L, \Delta \bar{E}^U] = [\Delta \bar{E}_W^L - \Delta \bar{E}_E^L, \Delta \bar{E}_W^U - \Delta \bar{E}_E^U]. \quad (22)$$

The membership and nonmembership degrees of the interval intuitionistic fuzzy expected return represent the acceptance and rejection of the strategy by the participants, respectively. Owing to the nature of a two-scale conflict, the maneuvering strategy should be ranked first [33]. Thus, the ranking algorithm based on the interval intuitionistic fuzzy entropy is presented in the following section.

V. STRATEGY OPTIMIZATION BASED ON FRACTIONAL-ORDER RNN MODEL

A. Fractional-Order RNN Model

To obtain the optimal expected return $[\Delta \bar{E}^L(p^w, p^e), \Delta \bar{E}^U(p^w, p^e)]$ in (22) for the mixed strategy (p^w, p^e) , the Pareto-optimal method [34] is used to realize the quantization of the interval-valued expected return (22). According to the Pareto-optimal method, the bi-objective program and the weighting problem corresponding to the program are, respectively, considered as follows:

$$\max_{p^w} \min_{p^e} [\Delta \bar{E}^L(p^w, p^e), \Delta \bar{E}^U(p^w, p^e)]$$

$$\text{s.t. } (p^w, p^e) \in (P^W, P^E). \quad (23)$$

$$\max_{p^w} \min_{p^e} \lambda_1 \Delta \bar{E}^L(p^w, p^e) + \lambda_2 \Delta \bar{E}^U(p^w, p^e)$$

$$\text{s.t. } (p^w, p^e) \in (P^W, P^E), \lambda_1 + \lambda_2 = 1, \lambda_1 > 0, \lambda_2 > 0. \quad (24)$$

If the solution (p^{w*}, p^{e*}) is optimal for (24), (p^{w*}, p^{e*}) is the Pareto optimal for (23), meaning that (p^{w*}, p^{e*}) is an efficient solution to (23).

Owing to the strong nonlinearity of the interval-valued expected return functions, the calculation of the optimization problem (24) becomes extremely complex. To simplify the calculation, the following modified expected return function is provided:

$$\max_{p^w} \min_{p^e} \sum_{i=1}^m \sum_{j=1}^n p_i^w (\lambda_1 a_{ij}^L + \lambda_2 a_{ij}^U) p_j^e$$

$$\text{s.t. } (p^w, p^e) \in (P^W, P^E), \lambda_1 + \lambda_2 = 1, \lambda_1 > 0, \lambda_2 > 0 \quad (25)$$

where $a_{ij}^L = \ln(1 - u_{ij}^{eL}) + \ln v_{ij}^{eL} - \ln(1 - u_{ij}^{wL}) - \ln v_{ij}^{wL}$ and $a_{ij}^U = \ln(1 - u_{ij}^{eU}) + \ln v_{ij}^{eU} - \ln(1 - u_{ij}^{wU}) - \ln v_{ij}^{wU}$. The obtained objective function (25) maintains the same monotonicity as (24). Subsequently, the optimization problem (23) is transformed into the following one:

$$\max_{p^w} \min_{p^e} p^w (\lambda_1 A^L + \lambda_2 A^U) p^e$$

$$\text{s.t. } (p^w, p^e) \in (P^W, P^E), \lambda_1 + \lambda_2 = 1, \lambda_1 > 0, \lambda_2 > 0 \quad (26)$$

where $A^L = \{a_{ij}^L\}$ and $A^U = \{a_{ij}^U\}$ are $m \times n$ -dimensional matrices.

According to the optimization problem (26), the following equation is obtained:

$$x^{*T} A y^* = \max_{x \in P^W} \min_{y \in P^E} x^T A y = \min_{y \in P^E} \max_{x \in P^W} x^T A y$$

where $A = \lambda_1 A^L + \lambda_2 A^U$. Then, the optimal strategies x^*, y^* that satisfy $x^{*T} A y^* = \max_{x \in P^W} \min_{y \in P^E} x^T A y =$

$\min_{y \in P^E} \max_{x \in P^W} x^T A y$ are equivalent to the following optimization problems [34], [35]:

$$(P) \begin{cases} \max c_y^T u \\ \text{s.t. } B_y^T u - A^L T x \leq 0 \\ B_y^T u - A^U T x \leq 0 \\ B_x^T x \leq c_x \\ u, x \geq 0 \end{cases} \quad (D) \begin{cases} \min c_x^T v \\ \text{s.t. } B_x v - A^L y \geq 0 \\ B_x v - A^U y \geq 0 \\ B_y y \geq c_y \\ v, y \geq 0. \end{cases}$$

Obviously, (P) and (D) constitute a pair of primal-dual linear optimization problems [35]. According to (P) and (D), the equivalent optimization problems are obtained as follows:

$$(P^*) \begin{cases} \max p^T \alpha \\ \text{s.t. } F \alpha \leq \mu, \end{cases} \quad (D^*) \begin{cases} \min q^T \beta \\ \text{s.t. } G \beta \leq \lambda \end{cases}$$

where

$$\alpha = \begin{bmatrix} u \\ x \end{bmatrix}, p = \begin{bmatrix} c_y \\ 0 \end{bmatrix}, F = \begin{bmatrix} B_y^T & -A^L T \\ B_y^T & -A^U T \\ 0 & B_x^T \\ -I & 0 \\ 0 & -I \end{bmatrix}, \mu = \begin{bmatrix} 0 \\ 0 \\ c_x \\ 0 \\ 0 \end{bmatrix}$$

$$\beta = \begin{bmatrix} v \\ y \end{bmatrix}, q = \begin{bmatrix} c_x \\ 0 \end{bmatrix}, G = \begin{bmatrix} -B_x & A^L \\ -B_x & A^U \\ 0 & -B_y \\ -I & 0 \\ 0 & -I \end{bmatrix}, \lambda = \begin{bmatrix} 0 \\ 0 \\ -c_y \\ 0 \\ 0 \end{bmatrix}.$$

To solve the above optimization problems (P^*) and (D^*) , the corresponding fractional-order RNN model is constructed as follows.

First, we construct the Lagrangian function

$$L(\beta, \omega) = q^T \beta + \omega^T (G\beta - \lambda).$$

Then, the Karush-Kuhn-Tucker (KKT) optimality conditions for problem (D^*) are given by

$$\omega \geq 0, G\beta - \lambda \leq 0, \omega^T (G\beta - \lambda) = 0, q + G^T \omega = 0. \quad (27)$$

It indicates that β^* is the optimal solution of (D^*) if $(\beta^*, \omega^*)^T$ is the solution of the KKT optimality conditions (27).

Based on conditions (27), the corresponding fractional-order RNN model is constructed as follows:

$${}_{t_0}^C D_t^\gamma z(t) = \theta \rho(z(t)), z(t_0) = z_0 \quad (28)$$

where $0 < \gamma < 2$ denotes the fractional order of the Caputo differential [25], $z = (\beta, \omega)^T \in R^N$ denotes the state, and $\theta > 0$ denotes the convergence rate

$$\rho(z) = \begin{bmatrix} -q - G^T(\omega + G\beta - \lambda)^+ \\ (\omega + G\beta - \lambda)^+ - \omega \end{bmatrix}$$

where $(\cdot)^+$ denotes $(x)^+ = \max(0, x)$.

Remark 1: The Riemann-Liouville definition, the Grunwald-Letnikov definition, and the Caputo definition are three commonly used definitions of fractional calculus. Owing to more practical physical meaning in initial values, the Caputo definition has more advantages in applications of engineering and computer science [36]. Thus, we used the Caputo definition in this study.

Definition 6: $z^* = (\beta^*, \omega^*)^T$ is the equilibrium point of the fractional-order RNN (28) if it satisfies

$$\rho(z^*) = 0.$$

Theorem 1: $z^* = (\beta^*, \omega^*)^T$ is the equilibrium point of the fractional-order RNN (28) if and only if $(\beta^*, \omega^*)^T$ is the solution of the KKT optimality conditions (27).

Proof: When $z^* = (\beta^*, \omega^*)^T$ is the equilibrium point of the fractional-order RNN (28), this results in $\rho(z^*) = 0$ and

$$-q - G^T(\omega^* + G\beta^* - \lambda)^+ = 0, (\omega^* + G\beta^* - \lambda)^+ - \omega^* = 0$$

which are equivalent to the KKT optimality conditions (27). Similarly, if $(\beta^*, \omega^*)^T$ is the solution of the KKT optimality conditions (27), this obviously implies that

$$-q - G^T(\omega^* + G\beta^* - \lambda)^+ = 0, (\omega^* + G\beta^* - \lambda)^+ - \omega^* = 0$$

which ensures that $z^* = (\beta^*, \omega^*)^T$ is the equilibrium point of the fractional-order RNN (28).

Remark 2: The equivalence between the KKT optimality conditions (27) and the fractional-order RNN (28) is proven in Theorem 1. Therefore, the equilibrium point of the fractional-order RNN (28) is the optimal solution of (D^*) and the optimization problem (26). Then, how to calculate the equilibrium point of the fractional-order RNN (28) remains the critical issue. Notably, the solutions of the fractional-order RNN (28) converge to its equilibrium point, if the fractional-order RNN (28) is stable. Then, the stable equilibrium point could be approximately calculated by a numerical solution within sufficient iterations. In this case, the stability of the fractional-order RNN (28) is essential in the proposed algorithm, which is discussed in the next section.

B. Stability Analysis of Fractional-Order RNN

According to Theorem 1 in Section V-A, the stability of the equilibrium point $z^* = (\beta^*, \omega^*)^T$ of the fractional-order RNN (28) is essential for the optimization problem (D^*) and (D) . Thus, the stability analysis of the fractional-order RNN, which can guarantee the convergence of the solutions of the fractional-order RNN, is discussed in this section. First, the existence and uniqueness of the solution of the fractional-order RNN (28) are analyzed in Lemma 1 and Theorem 2.

Lemma 1 [25]: The solution of a Caputo differential equation described by

$${}^C D_t^\gamma z(t) = g(z(t)), z(t_0) = z_0$$

exists and is unique, if the derivative function $g(z)$ is Lipschitz continuous, that is, $\|g(z_1) - g(z_2)\| \leq L\|z_1 - z_2\|$, where $L > 0$ and $z_1, z_2 \in \mathbb{R}^N$.

Theorem 2: The solution $z(t)$ of the fractional-order RNN (28) exists and is unique.

Proof: According to system (28), the derivative function $\theta\rho(z)$ is continuous and piecewise-smooth, which implies its Lipschitz continuity.

Using 1-norm $\|\cdot\|_1$ yields the following inequality for any $z_1 = (\beta_1, \omega_1)^T, z_2 = (\beta_2, \omega_2)^T \in \mathbb{R}^N$:

$$\begin{aligned} & \|\theta\rho(z_1) - \theta\rho(z_2)\|_1 \\ & \leq \theta(\|G\|_1 + 1)\|(\omega_1 + G\beta_1 - \lambda)^+ - (\omega_2 + G\beta_2 - \lambda)^+\|_1 \\ & \quad + \theta\|\omega_1 - \omega_2\|_1 \\ & \leq \theta(\|G\|_1 + 1) \sum_{i=1}^{N_\omega} |(\omega_1 + G\beta_1 - \lambda)_i^+ - (\omega_2 + G\beta_2 - \lambda)_i^+| \\ & \quad + \theta\|z_1 - z_2\|_1 \\ & \leq \theta(\|G\|_1 + 1) \sum_{i=1}^{N_\omega} |(\omega_1 + G\beta_1 - \lambda)_i - (\omega_2 + G\beta_2 - \lambda)_i| \\ & \quad + \theta\|z_1 - z_2\|_1 \\ & \leq \theta(\|G\|_1 + 1)\|\omega_1 - \omega_2 + G(\beta_1 - \beta_2)\|_1 + \theta\|z_1 - z_2\|_1 \\ & \leq \theta\left[(\|G\|_1 + 1)^2 + 1\right]\|z_1 - z_2\|_1 \end{aligned}$$

where N_ω denotes the dimension of ω .

Thus, the derivative function $\theta\rho(z)$ is Lipschitz continuous, such that the existence and uniqueness of the solution $z(t)$ of the fractional-order RNN (28) can be ensured based on Lemma 1. This completes the proof.

Because the solution $z(t)$ of the fractional-order RNN (28) exists and is unique, we discuss the stability of the equilibrium point z^* in the next section.

Definition 7: The equilibrium point z^* of the fractional-order RNN (28) is Lyapunov stable if, for any $\varepsilon > 0$, there exists a $\delta > 0$ such that $\|z(t) - z^*\| < \varepsilon$ for all $t \geq t_0$ and $\|z_0 - z^*\| < \delta$.

To realize Lyapunov stability, Lemma 2 is introduced in the following.

Lemma 2 [25]: The equilibrium point z^* of the fractional-order RNN (28) is globally Lyapunov stable if there exist the positive constants a_1, a_2, b_1 , and b_2 and a non-negative continuously differentiable function $V: \mathbb{R}^n \rightarrow \mathbb{R}^+$ that satisfies

$$\begin{cases} a_1\|z(t) - z^*\|^{b_1} \leq V(z) \leq a_2\|z(t) - z^*\|^{b_2} \\ {}^C D_t^\gamma V(z) \leq 0 \end{cases}$$

where $0 < \gamma \leq 1$.

The stability of the equilibrium point z^* of the fractional-order RNN (28) is analyzed in Theorem 3.

Theorem 3: The equilibrium point z^* of the fractional-order RNN (28) is Lyapunov stable when $0 < \gamma \leq 1$.

Proof: Construct the following Lyapunov function:

$$V(z) = \|z - z^*\|_2^2 + \|\rho(z)\|_2^2.$$

Function $\rho(z)$ can be rewritten as

$$\rho(z) = Mz + N$$

where

$$\begin{cases} M = \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}, N = \begin{bmatrix} -q \\ 0 \end{bmatrix}, \text{ if } \begin{bmatrix} G \\ I \end{bmatrix}^T z < \lambda \\ M = \begin{bmatrix} -G^T G & -G^T \\ G & 0 \end{bmatrix}, N = \begin{bmatrix} -q + G^T \lambda \\ -\lambda \end{bmatrix} \\ \text{if } \begin{bmatrix} G \\ I \end{bmatrix}^T z \geq \lambda. \end{cases}$$

Obviously, M and N are both bounded and continuous on z , and $M^T + M$ is negative semidefinite. Thus, $\rho(z)$ is Lipschitz continuous and satisfies

$$\|\rho(z_1) - \rho(z_2)\|_1 \leq \sigma \|z_1 - z_2\|_1$$

where σ denotes the bound of M , that is, $\|M\|_1 \leq \sigma$.

The above conditions imply that

$$\begin{aligned} {}^C_0 D_t^\gamma V(z) &\leq 2(z - z^*)^T {}^C_0 D_t^\gamma (z - z^*) + 2\rho(z)^T {}^C_0 D_t^\gamma \rho(z) \\ &= 2\theta(z - z^*)^T (\rho(z) - \rho(z^*)) + 2\rho(z)^T M {}^C_0 D_t^\gamma z \\ &= 2\theta(z - z^*)^T M(z - z^*) + 2\theta\rho(z)^T M\rho(z) \\ &= \theta(z - z^*)^T [M^T + M](z - z^*) \\ &\quad + \theta\rho(z)^T [M^T + M]\rho(z) \leq 0. \end{aligned}$$

Thus, the equilibrium point z^* of the fractional-order RNN (28) with $0 < \gamma \leq 1$ is stable. This completes the proof.

Remark 3: According to Theorem 3, the equilibrium point z^* of the fractional-order RNN (28) with $0 < \gamma \leq 1$ is Lyapunov stable, ensuring the convergence of the solutions of the fractional-order RNN. This means that the fractional-order RNN (28) with $0 < \gamma \leq 1$ is effective for the optimization problem (26) or (27). When $1 < \gamma < 2$, the Lyapunov stability is undetermined. Thus, $0 < \gamma \leq 1$ is a restricted constraint of the fractional-order RNN algorithm. However, numerical simulations for $1 < \gamma < 2$ case are still discussed to compare the optimization performances, among different orders γ . In this case, the optimal order γ should be analyzed and determined at the beginning of each optimization problem. This process is not automatic or adaptive in the current algorithm, which will be improved in future work.

Remark 4: When the fractional-order RNN in (28) is stable and $z(t)$ converges to the equilibrium point z^* , z^* can be approximately calculated by sufficient numerical iterations of $z(t)$ in the network of (28). Based on the KKT optimality conditions in (27), z^* contains the optimal solution p^{w*}, p^{e*} of the optimization problem in (26). To illustrate the optimization steps based on the fractional-order RNN, an example is listed in the following simulation section.

VI. SIMULATION

In this section, an example is considered to verify the effectiveness of the proposed algorithm. To compare the countermeasures performance, the maneuvering decision-making model using the interval intuitionistic fuzzy game theory is denoted by “W” and the max–min decision-making algorithm used during the multi-UUV countermeasures process is denoted by “E.” According to the maneuvering decision-making model of the multi-UUV countermeasure, the corresponding fractional-order RNN model was constructed and tested to analyze its applicability and effectiveness.

In (13), the i th UUVs in two multi-UUV systems of the underwater countermeasure are labeled “ W^i ” and “ E^i ,” and the maneuvering strategy sets S_{W^i} and S_{E^i} each comprising seven pure strategies. To reduce the computational load, the sets S_{W^i} and S_{E^i} of each UUV are simplified as

$$S_{W^i} = \{w_{i1}, w_{i2}, w_{i3}\}, S_{E^i} = \{e_{i1}, e_{i2}, e_{i3}\} \quad (29)$$

TABLE III
INITIAL CONDITIONS OF TWO MULTI-UUV SYSTEMS

UUV	Position	Speed	Yaw	Pitch
W^1	(−40,10,80)	2	−60°	5°
W^2	(−40,−10,80)	2	60°	−5°
E^1	(40,10,80)	2	128°	3°
E^2	(40,−10,80)	2	−128°	−3°

where w_{i1} and e_{i1} denote “speed change,” including “maintain the pace,” “speed up,” and “slow down”; w_{i2} and e_{i2} denote “turn around,” including “left turn” and “right turn”; and w_{i3} and e_{i3} denote “pitch,” including “pitch up” and “pitch down.” In this example, the absolute value of the cruising speed of UUVs in a straight line in a certain direction is 2 m/s, and the absolute value of the accelerated speed of moving in a straight line is 0.10 for UUVs on one side and 0.12 m/s² for UUVs on the other side. The absolute value of the minimum cruising speed is about 1 m/s. The accelerated speeds of the yaw and pitch angles are $\pm 30^\circ$ and $\pm 10^\circ$, respectively.

Each multi-UUV system has two UUV and maneuvering strategy sets S_W and S_E

$$S_W = \{w_1, \dots, w_9\}, S_E = \{e_1, \dots, e_9\} \quad (30)$$

where w_1 and e_1 denote (w_{11}, w_{21}) and (e_{11}, e_{21}) , respectively; w_2 and e_2 denote (w_{11}, w_{22}) and (e_{11}, e_{22}) , respectively; w_3 and e_3 denote (w_{11}, w_{23}) and (e_{11}, e_{23}) , respectively; w_4 and e_4 denote (w_{12}, w_{21}) and (e_{12}, e_{21}) , respectively; w_5 and e_5 denote (w_{12}, w_{22}) and (e_{12}, e_{22}) , respectively; w_6 and e_6 denote (w_{12}, w_{23}) and (e_{12}, e_{23}) , respectively; w_7 and e_7 denote (w_{13}, w_{21}) and (e_{13}, e_{21}) , respectively; w_8 and e_8 denote (w_{13}, w_{22}) and (e_{13}, e_{22}) , respectively; and w_9 and e_9 denote (w_{13}, w_{23}) and (e_{13}, e_{23}) , respectively. Thus, the mixed strategy spaces of “W” and “E” become

$$\begin{aligned} P^W &= \left\{ p^w \mid p^w = (p_1^w, \dots, p_9^w)^T \in \mathbb{R}^9, p_i^w \geq 0, \sum_{i=1}^9 p_i^w = 1, \right\} \\ P^E &= \left\{ p^e \mid p^e = (p_1^e, \dots, p_9^e)^T \in \mathbb{R}^9, p_j^e \geq 0, \sum_{j=1}^9 p_j^e = 1, \right\}. \end{aligned}$$

According to Table II and (16), the importance level matrix I is designed as follows:

$$I = \begin{bmatrix} 1 & 1.25 & 1.75 & 2 \\ 0.8 & 1 & 2 & 1.75 \\ 0.57 & 0.5 & 1 & 1 \\ 0.5 & 0.57 & 1 & 1 \end{bmatrix}.$$

Subsequently, the threat weights are obtained as follows:

$$t_1 = 0.3391, t_2 = 0.3137, t_3 = 0.1736, t_4 = 0.1736.$$

For two multi-UUV systems, the initial conditions of W^1 , W^2 and E^1 , E^2 are listed in Table III and the performance indices of restricted constraints are shown in Table IV. In the tables, “MA” represents “maximum acceleration,” “MD” represents “maximum deceleration,” “MVDA” represents “maximum variation of yaw angle,” and “MVPA” represents “maximum variation of pitch angle.”

TABLE IV
PERFORMANCE INDICES OF TWO MULTI-UUV SYSTEMS

UUV	MA	MD	MVDA	MVPA
W	0.12	0.12	30°	10°
E	0.10	0.10	30°	10°

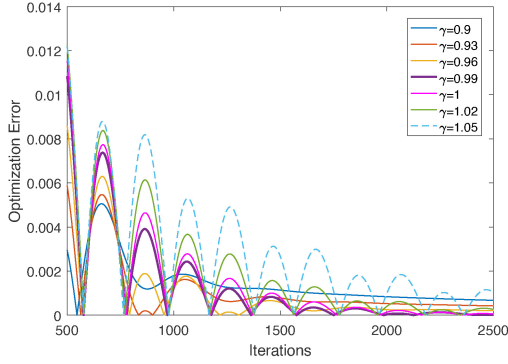


Fig. 2. Optimization errors with different fractional orders γ .

Denote the game step as k . The maximum game step is 40 ($k = 1, 2, \dots, 40$) and the interval time of each step is 5 s. Subsequently, based on (21)–(28) and the initial conditions of W_1 , W_2 and E_1 , E_2 , the corresponding fractional-order RNN model for the first step $k = 1$ can be constructed. Its parameters are obtained as indicated in Table IV.

To check the optimization performance, an optimization error ε is given based on (26)

$$\varepsilon = |p^w A p^e - p^{w*} A p^{e*}|$$

where $A = \lambda_1 A^L + \lambda_2 A^U$, p^w and p^e are the feasible solution obtained from (28), and p^{w*} and p^{e*} are the optimal solutions in (26). Obviously, a smaller optimization error ε (or closer to 0) means better optimization performance. According to Table V, for the first step $k = 1$, the states of the fractional-order RNN model and their optimization errors are calculated for different fractional orders γ , which are presented in Table VI and Fig. 2. When iteration is 1500, the optimal optimization error is 5.6518×10^{-4} with $\gamma = 0.97$. For more iterations, the optimization error becomes smaller. The computational performance of the proposed algorithm can realize 10^{-5} magnitude with 2000 iterations. In particular, the optimization error declines to 1.9019×10^{-5} with $\gamma = 0.99$ and 2500 iterations. Notably, the order parameter γ significantly influences the computational performance as shown in Fig. 2. The convergence rate of the smallest $\gamma = 0.9$ in Fig. 2 is the fastest, when the iteration is approximately less than 800. However, the convergence rate of the smallest $\gamma = 0.9$ declines with increasing iteration and becomes one of the worst choices after iteration 1300. Besides, the optimization error of $\gamma = 1.05$ is nonconvergent, meaning that a too large order γ may cause network instability. Thus, a smaller order γ is optimal for a small iteration, and a larger order γ is optimal for a large iteration. Based on Theorem 3, the fractional-order RNN is stable with $0 < \gamma \leq 1$, as verified in Fig. 2. According to Table VI and Fig. 2, when γ is close to 1, a smaller γ yields to a faster convergence rate; however, this results in a larger optimization error. Thus, $\gamma = 0.99$ is selected based on

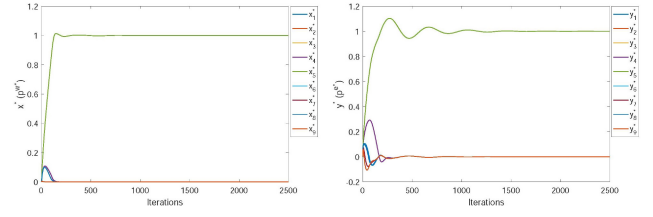


Fig. 3. States of fractional-order RNN model ($\gamma = 0.99$) in step $k = 1$.

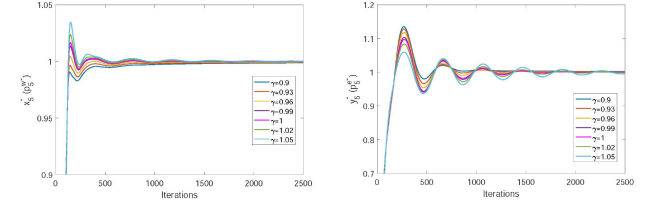


Fig. 4. States (x_5 and y_5) of fractional-order RNN model in step $k = 1$.

Table VI, such that it can ensure a sufficiently fast convergence rate and satisfy an adequately small optimization error.

As $\gamma = 0.99$ is used in the following steps, the equilibrium point $z^* = (\beta^*, \omega^*)^T$, which is the solution of the KKT optimality conditions (27), is calculated. As illustrated in Fig. 3, the states of fractional-order RNN converge to a certain equilibrium vector with sufficient iterations, such that the following optimal mixed strategies are obtained:

$$p^{w*} = x^* = (z_{12}^*, z_{13}^*, \dots, z_{20}^*)^T = (0, 0, 0, 0, 1, 0, 0, 0, 0)^T$$

$$p^{e*} = y^* = (z_3^*, z_4^*, \dots, z_{11}^*)^T = (0, 0, 0, 0, 0.9999, 0, 0, 0, 0)^T.$$

To verify the influence of the order γ , Fig. 4 depicts the states x_5 , y_5 with different orders γ . Clearly, $\gamma = 0.99$ is better in terms of both the convergence rate and optimization error.

According to the obtained mixed strategies, the strategy w_5 (w_{12}, w_{22}) is selected in probability. For “E,” e_5 (e_{12}, e_{22}) is selected based on the max–min decision-making algorithm. Subsequently, the states of both multi-UUV systems in the next step $k = 2$ can be calculated as follows.

UUV	Position	Speed	Yaw	Pitch
W^1	(−31.37, 5.02, 80.87)	2	-30°	5°
W^2	(−31.37, −5.02, 79.13)	2	30°	-5°
E^1	(30.74, 13.74, 80.52)	2	158°	3°
E^2	(30.74, −13.74, 79.48)	2	-158°	-3°

The above states are the initial conditions for the step $k = 2$, which can be used to calculate the corresponding parameters A^L and A^U and the fractional-order RNN model. The next optimal mixed strategy can be obtained based on the constructed fractional-order RNN, which gives the strategies of the next step $k = 3$ in probability. The above steps are repeated until k reaches the 40 maximum game steps.

The countermeasure processes between “W” and “E” are illustrated in Fig. 5, where the red dotted and solid lines represent “ W^1 ” and “ W^2 ,” respectively, and the blue dotted and solid lines represent “ E^1 ” and “ E^2 ,” respectively. In the first stage with steps $k \leq 11$, “ W^1 ” and “ W^2 ” attempt to attack

TABLE V
PARAMETERS OF FRACTIONAL-ORDER RNN MODEL IN THE FIRST STEP $k = 1$

$A^L =$	[	-0.0872	-0.1948	-0.0052	-0.1948	-0.3171	-0.1035	-0.0052	-0.1035	0.0706
		0.0521	-0.0537	0.1396	-0.0557	-0.1759	0.0416	0.1319	0.0351	0.2133
		-0.0920	-0.2033	-0.0091	-0.1962	-0.3222	-0.1062	-0.0065	-0.1060	0.0703
		0.0521	-0.0557	0.1319	-0.0537	-0.1759	0.0351	0.1396	0.0416	0.2133
		0.2159	0.1101	0.3018	0.1101	-0.0100	0.2053	0.3018	0.2053	0.3813
		0.0465	-0.0653	0.1306	-0.0555	-0.1818	0.0353	0.1348	0.0352	0.2127
		-0.0920	-0.1962	-0.0065	-0.2033	-0.3222	-0.1060	-0.0091	-0.1062	0.0703
		0.0465	-0.0555	0.1348	-0.0653	-0.1818	0.03520	0.1306	0.0353	0.2127
		-0.0969	-0.2047	-0.0104	-0.2047	-0.3272	-0.1088	-0.0104	-0.1088	0.0699
$A^U =$	[	0.0022	-0.0997	0.0802	-0.0997	-0.2157	-0.0129	0.0802	-0.0129	0.1523
		0.1418	0.0417	0.225	0.0396	-0.0743	0.1322	0.2178	0.1262	0.2952
		-0.0026	-0.1082	0.0773	-0.1012	-0.2207	-0.0145	0.0779	-0.0165	0.1519
		0.1418	0.0396	0.2178	0.0417	-0.0743	0.1262	0.225	0.1322	0.2952
		0.3059	0.2057	0.3877	0.2057	0.0918	0.2964	0.3877	0.2964	0.4635
		0.1363	0.0301	0.2175	0.0397	-0.0802	0.1275	0.2192	0.1247	0.2946
		-0.0026	-0.1012	0.0779	-0.1082	-0.2207	-0.0165	0.0773	-0.0145	0.1519
		0.1363	0.0397	0.2192	0.0301	-0.0802	0.1247	0.2175	0.1275	0.2946
		-0.0074	-0.1097	0.075	-0.1097	-0.2258	-0.0182	0.075	-0.0182	0.1515
$B_x =$	$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 \end{bmatrix}^T$	$c_x = [\begin{array}{cc} 1 & -1 \end{array}]^T$	$\lambda_1 = \lambda_2 = 0.5$,							
$B_y =$	$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 \end{bmatrix}$	$c_y = [\begin{array}{cc} 1 & -1 \end{array}]^T$	$\theta = 5$.							

TABLE VI
OPTIMIZATION ERROR OF FRACTIONAL-ORDER RNN
MODEL IN THE FIRST STEP $k = 1$

Order γ	Iteration 1500	Iteration 2000	Iteration 2500
0.9	1.1354e-03	8.3611e-04	6.7250e-04
0.91	1.0036e-03	7.2643e-04	5.8333e-04
0.92	8.8411e-04	6.2249e-04	4.9954e-04
0.93	7.7873e-04	5.2372e-04	4.2082e-04
0.94	6.9003e-04	4.2933e-04	3.4684e-04
0.95	6.2149e-04	3.3825e-04	2.7712e-04
0.96	5.7784e-04	2.4901e-04	2.1098e-04
0.97	5.6518e-04	1.5966e-04	1.4733e-04
0.98	5.9117e-04	6.7683e-05	8.4360e-05
0.99	6.6510e-04	3.0107e-05	1.9019e-05
1	7.9754e-04	1.3748e-04	5.3813e-05
1.01	9.9974e-04	2.5867e-04	1.4249e-04
1.02	1.2824e-03	3.9834e-04	2.6020e-04
1.03	1.6539e-03	5.6168e-04	4.2685e-04
1.04	2.1170e-03	7.5526e-04	6.7101e-04
1.05	2.6649e-03	9.8993e-04	1.0311e-03
1.06	3.2762e-03	1.2877e-03	1.5552e-03

“ E^1 ”; “ E^2 ” fails to chase “ W^2 ” and returns to the confrontation. Thereafter, “ E^1 ” escapes successfully in the second stage, and “ W^1 ” and “ W^2 ” quickly change their strategy and turn to attack “ E^2 ,” which is depicted in step $k = 14$ of Fig. 5. In the third stage, “ E^2 ” successfully escapes and attempts to attack “ W^1 ” with “ E^1 ,” as shown in step $k = 20$ of Fig. 5; “ W^2 ” rapidly turns back to the confrontation and attacks “ E^2 ,” which regains a favorable situation, as indicated in step $k = 24$ of Fig. 5. In the final stage, “ W^1 ” and “ W^2 ” reach the absolute advantage, and chase “ E^1 ” and “ E^2 ”; “ E^1 ” and “ E^2 ” escape together in the opposite direction to “ W^1 ” and “ W^2 ,” as illustrated in step $k = 34$ of Fig. 5. This example demonstrates the effectiveness of the obtained multi-UUV maneuvering decision-making algorithm as well as its superiority over the max-min decision-making algorithm.

Owing to the real-time requirements of the multi-UUV maneuvering decision making, the optimization calculation time in each step has to be less than the interval time

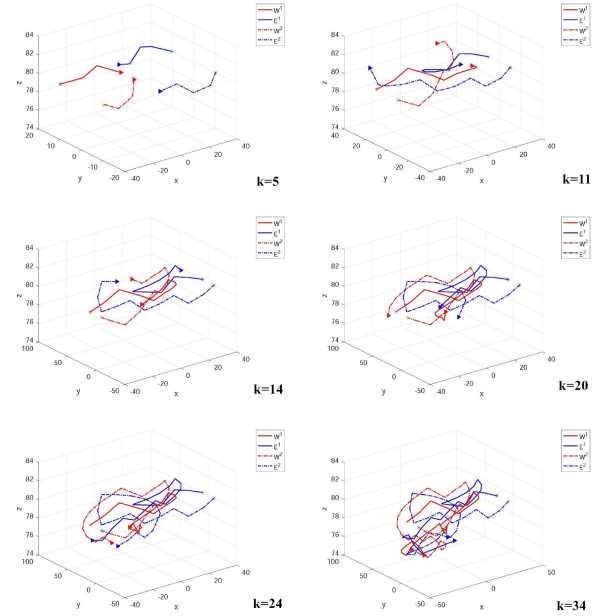


Fig. 5. Multi-UUV maneuvering counter-game in steps $k = 5, 11, 14, 20, 24$, and 34 .

of each step (5 s in the above example). Thus, the key point of the decision-making algorithms is how to solve the optimization problem in (26), with a smaller optimization error and shorter computing time. To compare the computational performance, some performance indices of fractional-order RNN, PSO, DE, and linear programming (LP) solver are listed in Table VII. Compared with LP, the proposed fractional-order RNN provides higher computation accuracy. Specifically, the optimization error of fractional-order RNN is reduced by 44.7% and the corresponding computation time is only increased by 4.5%. Compared with PSO and DE, the computation time of fractional-order RNN is much shorter, 2.1719 s. Moreover, the results also show that fractional-order RNN

TABLE VII
COMPUTATIONAL PERFORMANCE COMPARISON WITH OTHER DECISION-MAKING ALGORITHMS

Algorithms	Populations	Iterations	Optimization Errors	CPU Time (sec)
Fractional-Order RNN	1	2000	3.0107e-05	2.1719
PSO	100	300	3.2288e-05	3.0313
DE	100	300	5.1557e-05	3.6094
LP	none	12	5.4421e-05	2.0781

does not need rich populations and requires less computation than PSO and DE.

The above simulation test focuses on a certain case with initial conditions listed in Table V. The computation time rapidly increases as the number of participant agents increases. For example, when the numbers of the participant agents are both 10, the tested CPU time for reaching an error magnitude of $e-05$ is about 11 s. It will be unavailable if the cost time is larger than the set interval time in the decision making. Thus, the limits of the number of participants depend on the interval time of the decision making as well as the expected optimization error. How to balance the cost time and the number of participants will be considered and tested in future work.

VII. EXPERIMENT

In Section V, an example simulation of two multi-UUV systems of the underwater countermeasures is presented. In this section, the corresponding experiment on ground unmanned systems is presented to verify the effectiveness of the obtained results, algorithms, and simulations. The ground unmanned system countermeasure platform applied in this experiment mainly comprises four ground unmanned robots, a positioning and communication system with four base stations, and a central control computer. The ground unmanned robots are four-wheel omnidirectional differential-driven systems that we assembled. The computing system of the ground robots is manifold 2-G. The ultrawideband positioning technology is used for robot positioning. The robots network and communicate through a local area network router. The simulated communication cycle is set to 2 s, the maximum bit error rate is 10%, and the speed of the unmanned systems is 4 knots to effectively simulate the underwater environment. The central control computer is equipped with MATLAB 2016a and robot operating system (ROS). The calculation of the proposed countermeasure method is run by MATLAB including the LP solver of toolbox “CVX 3.0 beta,” and the interactions among the robots are achieved through ROS. The maneuvering performance limitations of these unmanned systems are set according to UUVs, but only the 2-D positions x and y in the last section are used.

Fig. 6 depicts the maneuver decision making of the unmanned systems in steps $k = 5, 11, 14, 20, 24$, and 34 , respectively. The green and blue arrows indicate the forward directions to simulate “W” and “E,” respectively, in Section V. The maneuver trends of the experiment on the unmanned systems are similar to the multi-UUV simulation results, which coincide with the obtained results and algorithms.

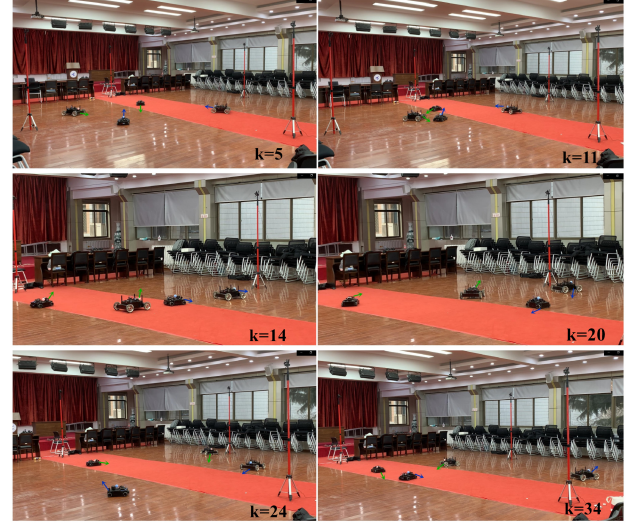


Fig. 6. Unmanned systems maneuvering counter-game in steps $k = 5, 11, 14, 20, 24$, and 34 .

VIII. CONCLUSION

In this study, a multi-UUV maneuvering counter-game algorithm for a dynamic target scenario was proposed using a fractional-order RNN. The key findings of this study focus on the interval-valued intuitionistic fuzzy mapping in multi-UUV counter-game model and fractional-order RNN in strategy optimization. The interval-valued intuitionistic fuzzy mapping was used to construct the maneuvering decision-making optimization model of the multi-UUV counter-game, which effectively describes underwater environments with weak connectivity, underwater noise, and dynamic uncertainties. Moreover, a fractional-order RNN was constructed based on the KKT optimality conditions of strategy optimization, which provides an improvement in convergence rate and optimization error by the tunable fractional-order parameter. The optimal strategy was designed as the equilibrium point of the fractional-order RNN, whose stability was confirmed to ensure the convergence of the solutions of the fractional-order RNN. Further, an example of a multi-UUV dynamic counter-game and an experiment were developed using unmanned ground systems were presented herein. A comparison of both examples demonstrated that the proposed algorithm was more effective than and superior to the commonly used max–min algorithm in multi-UUV dynamic counter-games. The order of the fractional-order RNN, as well as the choice of the optimal maneuvering strategy, was also observed to influence the convergence rate and optimization error of the proposed algorithm. Our future work shall focus on considering different communication conditions and formation types, which

will bring about more constraints and computation burdens in the game process. In addition, how to balance the computation time of the optimization algorithm and the number of participants will be considered.

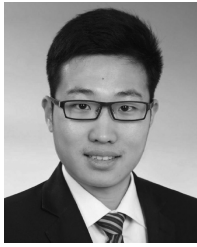
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